Lab 6: Radiometer

Measure how illumination and thermal history affect a sealed rotor. Report due: Monday, December 7, 11:59 PM. Complete Pre-Lab 6 including Q6 before your section.

What you will measure

Record repeatable vane passages in video. Calculate revolutions and rpm, compare matched observation windows at different measured lux levels, and check early/late speed changes. Pressure and vane temperature inside the sealed bulb are not measured.



Course apparatus: glass bulb, rotating vanes and a lux meter.

1. Setup and quick feasibility check

Use the radiometer, lamp, lux meter, ruler and a fixed camera. Keep the bulb upright, the lamp setting fixed and the background outside the strongest light beam. Use instructor-approved lamp clearance. Measure lux beside the rotor plane at the same meter position and orientation each time.

Lock camera exposure, gain and focus. Save the actual capture fps from the camera settings, not the exported movie playback rate. Record a short preview, watch a complete revolution and verify the event count p per turn. A unique vane crossing gives p = 1; other event definitions require a full-turn check.

Use enough physical duration to see several passages. A high-speed rate is not automatically helpful for a slow rotor. If camera memory is too short, use a known-rate continuous phone video. Confirm a usable recording before collecting all runs.

2. Collect four core runs

A1: reference illuminance. B1 and C1: two other safe lamp distances, same lamp setting. A2: repeat reference A after comparable cooling. The instructor selects workable distances during the bench check; record actual distance and lux for every run.

For each run: (1) let the rotor and apparatus return to a comparable initial condition; (2) start video before switching on the lamp; (3) record the illumination-on frame; (4) film the common 60–90 s window, or a longer common window if the motion is slow; (5) save the original movie and log. Aim for at least six consecutive verified event intervals. If that is not feasible, extend the window or reduce the number of conditions with the GTA.

Record cooling time and whether the rotor has stopped. Keep camera, meter and lamp orientation fixed. Start A1, B1, C1 and A2 comparably; a warm A is a separate thermal-history condition, not an independent cool-start repeat.

3. Mark crossings and obtain rpm

Unzip the MATLAB bundle. Run Mark_Lab6_Passage_Frames.m. Right/Left navigate; Up/Down change the step; Space marks; Backspace undoes; Q saves. At each crossing use a one-frame step. Mark the same event and direction in increasing original frame numbers. Frame zero is the first frame.

Run Run_Lab6_Radiometer.m, select the saved passage table, and enter capture fps and verified p. Inspect cumulative revolutions and interval rpm. Review flagged intervals in the film. Never silently interpolate a missed passage or differentiate a noisy wrapped angle.

Timing equations

$t = (n - n_{on})/f_c$, where n is original frame index, n_{on} is illumination-on frame and f_c is actual capture fps. For ΔN consecutive event intervals spanning Δt seconds, $rpm = 60\,\Delta N/(p\,\Delta t)$. Number of intervals equals number of marks minus one. Angular speed [rad/s] = $2\pi \times rpm/60$.

Optional: thermal memory

After an illuminated run, block the light without moving the bulb and continue filming. Note the blocking frame and whether rotation persists or changes. A warm repeat of A can reveal history effects. Remaining thermal gradients may still drive the rotor; light-off motion is not automatically a pure viscous spin-down.

4. Compare the illumination runs

Fill Lab6_Comparison_Log.csv: run ID, passage CSV filename, capture fps, verified p, lux, distance [m], illumination-on frame, requested window start/end [s], and initial state. Passage file paths can be relative to the log folder. Use the same requested physical window for matched comparisons.

Run Compare_Lab6_Illumination_Runs.m. It exports interval rpm versus time since light-on, mean rpm versus measured lux, and a summary with early/late means and covered times. Means cover the first-to-last retained crossing; unobserved edges of a requested window are not invented. Call a response steady only if the time trace supports it.

5. Report figures and discussion

Include: (1) labeled setup photo and settings; (2) cumulative turns and interval rpm for one run; (3) matched-window mean rpm versus measured lux with reference repeats labeled; (4) early/late comparison and an explanation of thermal history. Include the light-off response if collected. Export figures directly.

The rotational balance is $I\,d\omega/dt = \tau_{gas}(t) - \tau_{loss}(\omega)$. I is rotor inertia [kg·m²], ω is angular speed [rad/s], and torque is in N·m. An exponential approach to a final speed assumes constant gas torque and linear loss bω. Heating can violate the constant-torque assumption. A fitted time scale need not be purely mechanical.

Lux is weighted for human vision, not irradiance in W/m². We do not know the bulb pressure, vane temperatures, inertia or calibrated loss torque. Do not infer exact Knudsen number, torque, optical efficiency or absorbed power from rpm and lux alone.

Required molecular activity

Complete Q6 in the Pre-Lab 6 notebook before the laboratory. Run the molecular sampler, compare with the analytical wall model, and explain temperature contrast and sample count. Use the molecular-force figure to interpret your measured direction and early/late speed changes in the report. The wall model does not quantitatively predict the bulb rpm.

Before leaving

Show the GTA a usable movie, capture setting, verified p, one passage analysis and the run-comparison figure. Keep raw movies, passage CSVs, the completed log and MATLAB exports. All group members should have access to the files.

Research connections: Baier, Hardt, Shahabi & Roohi (2017); Lotfian & Roohi (2018).